Latent Spaces for Symbolic Music (MIDI) via Piano-Roll and VAE/Diffusion Pipelines

Executive overview

This report surveys representation schemes, latent-variable models, diffusion pipelines, multi-modal extensions, and evaluation methodologies relevant to building a bar-level, per-track latent space for symbolic music (MIDI), with a focus on piano-roll-as-image front ends and VAE-based latents suitable for downstream diffusion.
It synthesizes work from symbolic music VAEs, piano-roll generative models, latent diffusion for scores, and controllable/disentangled representation learning, and also discusses how to **reuse general-purpose image VAEs** (such as those from Stable Diffusion and SDXL) as front-end encoders for bar-level piano-roll images.[1][2][3]

Problem framing and design goals

For the target system, each bar of each track (or small group of tracks) should be encoded to one latent vector, forming a sequence of bar-level embeddings per track that can be used both for reconstruction and as the domain for a symbolic or latent diffusion model.
Work such as MusicVAE, Hierarchical-EC2-VAE, MeasureVAE, MuseMorphose, and PhraseVAE already explore bar-/phrase-level latents and provide architectural patterns for such representations.[2][4][5][6][7]

Key design goals inferred from prior work:

- **Local fidelity**: Bar-level latents must preserve timing, pitch content, and basic voice-leading within each bar.[8][2]

- **Global structure**: Latent sequences over bars should support long-term structure modeling (e.g., via hierarchical decoders or diffusion over sequences).[4][9][2]
- **Controllability**: Latent dimensions or learned directions should support interpretable edits (density, rhythm, style, instrumentation, groove, etc.).[10][5][6][11]
- **Compatibility with diffusion**: Latent vectors should have approximately Gaussian priors (for continuous diffusions) or discrete codebooks (for discrete diffusion) and be compact enough for efficient iterative sampling.[3][9][12]

Symbolic representations: piano rolls and beyond

Piano-roll representation and datasets

A piano roll is a matrix with pitch on the vertical axis, time on the horizontal axis, and entries representing note activity, commonly velocity.[13]
The Lakh Pianoroll Dataset (LPD) encodes 4/4 bars at 24 time steps per beat (96 steps per bar) and 128 MIDI pitches, producing a 96×128 matrix per bar and per track, with multi-track sequences represented as 96×128×M tensors.[14][13]

Pypianoroll is a widely used Python library that parses MIDI to multitrack piano rolls, supports binary or velocity-valued matrices, and provides utilities for manipulation, visualization, and evaluation.[15][16][17]
MuseGAN and MIDI-Sandwich2 use LPD-style multi-track rolls (often merged into 5 or 17 instrument categories) as inputs for deep generative models, showing the feasibility of multitrack pianorolls as a standard representation.[18][19][14]

Augmented piano rolls and image-like encodings

Several VAEs and related models extend basic piano rolls with extra channels or structured layouts:

- **MIDI-VAE** augments standard pitch piano rolls with separate velocity and instrument rolls, and uses three encoders/decoders

with a shared latent to jointly model pitch, dynamics, and instrumentation.[20][21][22]
- **SCG latent diffusion** encodes segments of piano rolls with 3 channels (velocity, onset roll, and pedal roll) into 3×128×128 tensors for a VAE, which then feeds a DiT-based latent diffusion model.[23][3]
- A tonal-tension VAE uses 64×89 piano-roll inputs (melody+bass) with multiple feature outputs to learn latent directions corresponding to tonal tension measures, indicating utility of multi-headed decoders and auxiliary channels.[24]

From these, a bar-level image tensor for each track or track-group could reasonably be:

- Size: H = 128 pitches, W corresponding to temporal resolution (e.g., 32, 48, 64, 96, or 128 columns per bar), often a power of 2 when using image VAEs.[3][13]
- Channels: at least 1–3 channels: binary on/off, velocity, and possibly onset-only masks; pedal and articulation channels are useful for piano or expressive data.[20][3]
- Track axis: either separate images per track or track-as-channel for limited, fixed instrument sets (e.g., LPD-5's drums/piano/guitar/bass/strings).[19][14]

These choices allow reuse of off-the-shelf **image VAEs** or LDM-style encoders while retaining musically salient structure.

Symbolic- and music-specific VAEs vs. generic image VAEs

Music-specific latent encoders (summary)

Music-specific VAEs such as MusicVAE, EC2-VAE, MuseMorphose, MIDI-VAE, GrooVAE, and PhraseVAE/PhraseLDM are trained directly on symbolic music sequences or piano-rolls and produce latents with clear musical semantics (bars, phrases, grooves, style attributes).[11][25][26][7][2][3][20]
These encoders typically:

- Operate on MIDI/event sequences or custom structures (PianoTree, chord graphs) rather than raw images.[6][8]
- Include musically motivated factorization (pitch/rhythm, score/performance, instrument channels) and losses (e.g., tonal tension or groove targets).[27][24][20]
- Are relatively small and trained on music-only corpora, so their inductive biases and priors are tightly matched to symbolic music.

They are a strong baseline if one is willing to train encoders from scratch or fine-tune them and if interoperability with image diffusion ecosystems is not a requirement.

Generic image VAEs from diffusion models

General-purpose latent diffusion models such as Stable Diffusion and Stable Audio use a **two-stage architecture**: a VAE that maps input data to a low-dimensional latent tensor and a diffusion model that operates in this latent space.[28][29][1]
For Stable Diffusion v1.x, the VAE compresses RGB images (H×W×3) down to a latent tensor with 4 channels and a spatial downsampling factor of 8 (i.e., 512×512 images map to 64×64×4 latents).[30][31]
SDXL, SD-VAEs such as `sd-vae-ft-mse`, and related variants (eq-vae, Playground, FLUX, etc.) follow similar designs with different channel counts, downsampling factors, and training data, but they all provide a **pretrained, high-capacity image autoencoder** with approximately Gaussian priors, designed for compatibility with diffusion.[31][30]

In the audio domain, Stable Audio shows that a Stable-Diffusion-like VAE can be repurposed to compress 44.1 kHz stereo waveforms into low-frequency latents with strided convolutions and multi-resolution STFT losses, confirming that such VAEs can generalize to **non-image modalities** when trained appropriately.[29][32][1]

The question here is whether one can take *off-the-shelf* image VAEs (e.g., from Stable Diffusion v1.4, SDXL, FLUX) and use them as **frozen encoders/decoders** for bar-level piano-roll images, instead of training MIDI-specific VAEs.

Using general-purpose image VAEs as piano-roll encoders

Mapping piano-roll bars into the image VAE input space

Since Stable Diffusion-type VAEs are trained on 3-channel images, piano-roll tensors must be embedded into this input space:

- **Channel mapping**: Map one to three musically meaningful channels (e.g., velocity, onset mask, pedal) into RGB channels, possibly using simple linear remappings (velocity → luminance, onset mask → chroma channel, pedal → blue channel).[3][20]
- **Spatial layout**: Treat pitch as one spatial dimension and time as the other. For a 4/4 bar with 96 time steps and 128 pitches, this yields a 128×96 "image"; pad or resample to a resolution compatible with the chosen VAE (e.g., 128×128 or 256×256), preserving pitch/time orientation.[13][3]
- **Normalization**: Normalize velocities and other channels into the or [−1,1] range expected by the image VAE, and ensure that empty bars and dense bars map to clearly distinct patterns.[10]

Once this mapping is fixed, any bar-track (or bar-group) can be rasterized and passed through a pretrained image encoder such as:

- Stable Diffusion v1.x VAE (`stable-diffusion-v1-4` encoder)
- SDXL VAE (`sdxl-vae`, `eq-sdxl-vae`)
- Fine-tuned SD VAEs (`sd-vae-ft-mse`, `eq-vae-ema`)
- Newer, compact autoencoders (e.g., FLUX Tiny AutoEncoder, Playground VAE variants)

which produce compact latent tensors suitable for diffusion, typically with 4–16 channels and aggressive spatial downsampling.[30][31]

Advantages of using pretrained image VAEs

Using these pretrained image VAEs offers several concrete benefits:

- **No VAE training cost**: The expensive stage of learning a good compression model is already done; only the diffusion model over latents and downstream heads need to be trained.

- **High capacity and robustness**: These VAEs are trained on massive, diverse image datasets and typically capture rich spatial patterns and are robust to noise and distributional variation.[31][30]
- **Immediate compatibility with diffusion backbones**: Latents are already in the format expected by DiT/UNet diffusion models used in Stable Diffusion variants; this mirrors how Stable Audio leverages a Stable-Diffusion-like autoencoder for audio.[1][29]
- **Engineering convenience**: Existing codebases and checkpoints support fast encoding/decoding (tens of microseconds per patch on modern GPUs), which aligns with the throughput numbers observed for models like `sd-vae-ft-mse`, `eq-vae-ema`, SDXL-VAEs, and FLUX autoencoders.

These properties make generic image VAEs an attractive drop-in for bar-level piano-roll images, especially when many different diffusion architectures (2D, DiT, recurrent) are to be tried over the same latent grid.

Limitations and domain mismatch

However, several caveats arise from the domain mismatch:

- **Perceptual loss mismatch**: Image VAEs are trained with image-oriented reconstruction losses (pixel MSE, perceptual losses in image feature space), not music-specific objectives like onset timing, voice-leading, or tonal tension, so fine-grained musical features may be reconstructed poorly even when visual similarity is high.[33][3]
- **Semantic bias**: The latent space is biased toward natural image statistics, which might compress or discard structures common in piano rolls (e.g., vertical stripes for chords, highly regular grids) differently from an audio- or MIDI-trained encoder.[30][13]
- **Channel semantics**: RGB channels in images carry correlated color semantics, whereas piano-roll channels (velocity, onset, pedal) have distinct roles; mapping them naïvely may make some musically important signals hard for the VAE to model.
- **Quantization and binarization**: For binary piano rolls, continuous-valued reconstructions from the image VAE must be

thresholded or decoded via separate networks, potentially introducing artifacts in note on/off boundaries.[34]

These trade-offs suggest that while off-the-shelf image VAEs are attractive for rapid experimentation and to leverage existing diffusion tooling, fine-tuning or music-specific VAEs may still be needed for best musical fidelity.

Hybrid strategies: frozen vs. fine-tuned VAEs

A pragmatic approach is to start with a frozen image VAE and then gradually introduce music-specific adaptations:

- **Frozen encoder, learned post-decoder**: Use the pretrained encoder to obtain latents for each bar-image, but discard the original decoder. Instead, train a MIDI-specific decoder that maps latents directly to symbolic outputs (e.g., event sequences or piano-rolls), allowing the encoder to remain visual while the decoder learns a musically aligned inverse.
- **Fine-tuning the VAE on piano-roll images**: As in Stable Audio, where the audio VAE is trained/fine-tuned with STFT-based losses over audio, one can fine-tune an image VAE on piano-roll datasets with music-specific reconstruction losses (onset F1, velocity RMSE, chord accuracy), gradually aligning the latent manifold with musical semantics while preserving its compatibility with diffusion.[29][1]
- **Multi-task or dual-head decoders**: Keep the encoder fixed and train dual decoders: one that reconstructs the piano-roll image (for visual inspection and compatibility) and another that outputs symbolic tokens or high-level features (chords, rhythm patterns), similar to tonal-tension VAEs with multiple output heads.[24]

These hybrids exploit the availability and efficiency of general-purpose image encoders while progressively injecting music-specific inductive biases.

Latent-variable models for symbolic music

MusicVAE and hierarchical RNN VAEs

MusicVAE introduces a hierarchical latent vector model using a bidirectional LSTM encoder and a hierarchical decoder that outputs embeddings for subsequences (e.g., bars) which are then decoded into note sequences.[35][2][4]
For short 2-bar sequences, it uses a flat RNN VAE; for 16-bar monophonic and trio models, it introduces a conductor RNN whose steps correspond to groups of bars, encouraging the model to use the latent code over long timescales.[2][35]
MusicVAE supports attribute-vector arithmetic (e.g., a "note density" vector) and interpolation in latent space, providing an early demonstration of semantically meaningful operations on musical latents.[10][2]

Key relevance for bar-level latents:

- The hierarchical decoder effectively learns a sequence of "bar embeddings"; these can be interpreted as bar-level latents for downstream models, even if not explicitly exposed.
- The method tackles posterior collapse via hierarchical decoding and KL warm-up / free-bits, which are essential design points when compressing full bars into one vector.[2]

PianoTree VAE and tree-structured latents

PianoTree VAE argues that naive piano-roll or MIDI-like encodings are suboptimal for polyphony and proposes a tree-structured representation (PianoTree) with a VAE that learns semantically meaningful latents for polyphonic segments.[8]
The model achieves better reconstruction and latent geometry compared to conventional piano-roll VAEs, supporting polyphonic chords and voicing.[8]

Although its encoding is not image-based, it provides evidence that structured, musically informed decoders (e.g., representing chords explicitly) can significantly improve latent quality for polyphony, a consideration when choosing between purely image-based or event/tree representations.

EC2-VAE and hierarchical EC2-VAE

EC2-VAE learns disentangled latent representations for controllable polyphonic music generation by factorizing the latent into pitch and rhythm subspaces and pairing one part with a rhythm decoder.[36][27]
Hierarchical-EC2-VAE extends this to longer phrases by stacking multiple EC2-VAEs and using an additional hierarchical predictive layer, yielding phrase-level latents that decode to bar-level and then note-level representations.[27]

These works demonstrate:

- Effective disentanglement of rhythm vs pitch in VAEs via architectural factorization and appropriate regularization.[27]
- Hierarchical bar/phrase representations that can be manipulated at different temporal scales, similar to a bar-wise or phrase-wise latent design.

MeasureVAE and adversarialVAE

MeasureVAE focuses explicitly on **measure-level** representations of music across genres, comparing it with an AdversarialVAE variant.[37][5]
Experiments show that MeasureVAE achieves better reconstruction while AdversarialVAE provides stronger independence among musical attributes; for 4 regularized latent dimensions, 32–64 total latent dimensions are recommended as an optimal trade-off across genres such as pop and rock.[5]

MeasureVAE's design is directly aligned with "one latent per bar" and offers empirical guidance on latent dimensionality, attribute regularization, and genre robustness.

MuseMorphose and bar-wise Transformer VAE

MuseMorphose is a Transformer-based VAE for style transfer of 32-bar pop piano performances, using an encoder that operates **bar-wise** to extract a latent "blueprint" for each bar, with attributes like rhythmic intensity and polyphony controlled at the bar level.[38][6]
The decoder is a Transformer with in-attention conditioning, trained

with a VAE objective to enable bar-level style transfer with user-specified attribute sequences.[6]

This work provides:

- A concrete precedent for **bar-level latent vectors** governing generation.
- A demonstration that bar-latents can encode expressive performances with timing and dynamics, not just quantized notes.[6]

Drum and groove VAEs

GrooVAE (and related models like PocketVAE) target drum/groove modeling by encoding sequences of drum hits with both score (which drums) and groove (microtiming and velocity) into a latent space.[39][25][11]
GrooVAE jointly encodes quantized MIDI, per-note velocity, and microtiming offsets into a shared latent, and demonstrates attribute-like controls such as adding groove to a rigid score or infilling missing kit elements.[11][39]
PocketVAE uses a discrete latent representation (VQ-VAE) for drum notes and separate continuous latents for velocity and microtiming, enabling controllable drum groove generation from templates.[25]

These models illustrate how bar-length (or phrase-length) rhythmic patterns can be compressed into latents that disentangle discrete note content from expressive timing, with straightforward parallels to per-bar track latents.

Classifying VAEs and conditional VAEs

Classifying VAEs add a discrete class variable to the VAE for tasks like generating music in specific keys, demonstrating improvements in staying in key compared to standard VAEs trained on untransposed data.[40]
MIDI-VAE, beyond multi-channel rolls, uses style labels for style transfer between genres (e.g., classical vs jazz), showing how style conditioning can be combined with shared continuous latents for pitch/velocity/instrument.[41][22][20]

Such conditional structures are natural extensions if bar-level latents need to carry instrument class, style, or role (melody/bass/drums) in addition to raw notes.

Latent diffusion and diffusion-over-latents for symbolic music

Diffusion over sequence latents (MusicVAE + diffusion)

Mittal et al. propose training diffusion models on sequences of latent embeddings obtained from a pre-trained VAE, avoiding direct diffusion on discrete symbol sequences.[42][9]
For symbolic music, they encode tokens (from a MusicVAE-like model) into continuous embeddings and train a non-autoregressive diffusion model over these sequences, achieving strong unconditional generation and infilling compared to autoregressive LMs.[9][12]

This architecture is directly aligned with using a bar-level encoder as a front end and then applying diffusion over the resulting sequence of bar latents.

Latent diffusion over piano-roll images (SCG)

Huang et al. (SCG) present a latent diffusion architecture for symbolic music, where a VAE encodes 3×128×128 piano-roll segments into latents, and a DiT-based diffusion model is trained in this latent space.[23][3]
The paper shows that this approach can generate high-time-resolution (10 ms) performances with velocity and pedal information, and that latent diffusion outperforms pixel-space diffusion baselines.[3]

Additionally, SCG introduces **stochastic control guidance** (SCG) for non-differentiable rule-guided diffusion, enabling plug-and-play conditioning on rule-based metrics such as note density or chord progression.[43][3]
For a bar-level image-VAEs pipeline, SCG's architecture is the closest reference: it already answers how to shape the VAE encoder/decoder, how big each piano-roll image should be, and how to couple the VAE with a transformer diffusion backend.

In the proposed adaptation, SCG's custom VAE could be replaced by an SD/SDXL-type pretrained VAE operating on piano-roll images, with minimal changes to the diffusion backend.

Phrase-level latent diffusion (PhraseVAE / PhraseLDM)

PhraseVAE / PhraseLDM propose a full-song symbolic generation framework in which a PhraseVAE compresses a variable-length polyphonic note sequence (phrase) into a 64D latent, and PhraseLDM performs latent diffusion over phrase-level latents rather than note tokens.[26][7]
The authors explore multiple compression strategies, including direct phrase compression, bar-based compression, and a **PositionVAE** that produces one latent per active temporal position inside a bar.[44][7]

Key lessons for bar-level design:

- There is a trade-off between compactness and reconstruction fidelity; more compact latents make accurate reconstruction harder but simplify diffusion, while finer-grained latents (e.g., position-wise) improve fidelity at the cost of sequence length.
- Phrase-level diffusion can generate long pieces (up to 128 bars) without autoregression, suggesting that bar-latent sequences are a tractable domain for LDMs.[7][26]

Discrete and symbolic diffusion variants

Several works apply diffusion directly to piano-rolls or symbolic tokens:

- Early symbolic diffusion models parameterize discrete sequences in the continuous latent space of a pre-trained VAE, as in Mittal et al.[42][9]
- D3PIA uses discrete diffusion for piano accompaniment, treating piano-roll pixels as discrete note states or operating in VQ-VAE latent space.[45]
- Discrete diffusion probabilistic models for symbolic music train DDPMs over discrete sequences, sometimes with MusicVAE latents as priors for long-term structure.[12][46]

While these do not always use image VAEs, they show viable alternatives if discrete latent codes (VQ-VAE) or masked-token diffusion are desired instead of continuous Gaussian latents.

Disentanglement, attribute control, and structured latent spaces

Attribute vectors and feature disentanglement

MusicVAE shows that linear directions in latent space correspond to interpretable musical attributes, such as note density, enabling attribute vector arithmetic (e.g., adding a "more notes" vector).[10][2]
A tonal-tension VAE learns latent vectors that correspond to changes in tonal tension, validated via external measures (tensile strain and cloud diameter), and shows that multi-output VAEs with auxiliary supervised heads can identify controllable latent directions.[24]

EC2-VAE and Hierarchical-EC2-VAE disentangle pitch and rhythm factors and demonstrate that bar-level representations can maintain disentanglement across longer sequences.[36][27]
MeasureVAE and AdversarialVAE compare reconstruction vs attribute-independence trade-offs across latent dimensionalities and attribute-regularization setups, concluding that moderate latent sizes (32–64) strike a good balance for interpretable control across genres.[5]

A related line of work sets up latent-variable models with Gaussian mixture priors and semi-supervised clustering to learn high-level attributes (e.g., arousal) on top of low-level latent features, providing a framework for building "faders" that independently adjust different musical aspects.[47]

Discrete latent codes (VQ-VAE) and bar-level codes

PocketVAE models drum notes via a vector-quantized VAE (VQ-VAE) for discrete note patterns combined with continuous latents for velocity and microtiming, showing benefits of discrete codes for capturing categorical rhythmic patterns.[25]
Other rhythm-oriented work (e.g., MIDI-Sandwich2) uses hierarchical VAEs and improved binarization for MuseGAN-style piano-rolls,

indicating that discrete representations can be beneficial for robust multi-track structure modeling.[18]

Discrete codes are compatible with discrete diffusion (as in D3PIA and other DDPM variants) and can yield compact per-bar indices that are amenable to symbolic editing and retrieval.[45][12]

Multi-modal and feature-augmented latent spaces

Instrumentation, dynamics, and performance features

MIDI-VAE extends standard piano-rolls by including separate velocity and instrument channels and training shared-latent encoders/decoders for pitch, velocity, and instrument, enabling style transfer across pitch, dynamics, and timbre.[21][22][20]
GrooVAE and related models provide explicit control over performance attributes such as groove (timing deviations and velocities), either by encoding both score and groove in the same latent or by predicting missing components given one of them.[39][11]

These ideas generalize naturally to a bar-level image-VAEs framework: one can add channels for velocities, articulations, instrument IDs or embeddings, or performance controls, either as additional pixel-wise channels or as side-condition vectors concatenated to latents.

Audio-symbolic joint embeddings

Cross-modal representation learning has been applied to map audio and symbolic representations into shared latent spaces:

- A cross-modal framework for audio-to-symbolic arrangement learns a latent representation from both audio (mel-spectrograms) and symbolic data, then decodes to MIDI, enabling audio-conditioned arrangement and style transfer; code and demos are available.[48]
- JASCO combines symbolic conditions (e.g., chords and melody) with audio-based conditions (e.g., separated drums or full mix)

inside a text-to-music generative model, using bottleneck layers to fuse conditions with text and enabling both local and global control.[49]
- Recent multi-modal music synthesis systems such as MIAO explore neural MIDI-to-audio synthesis with joint consideration of MIDI and audio, enabling precise note-level control while handling multiple instruments, and rely on learned joint embeddings to align MIDI and audio features.[50][51]

Such joint embeddings could be integrated as:

- Side-conditioning vectors attached to bar-level symbolic latents (e.g., per-bar audio summary or timbral embedding).
- A shared latent space where both MIDI-derived bar-latents and audio-derived embeddings live, facilitating audio-conditioned symbolic diffusion.

Cross-modal and multi-source diffusion

Multi-source latent diffusion models in the audio domain demonstrate that separate source latents (one per instrument) can be concatenated and modeled jointly in a diffusion model to produce coherent multi-track mixtures and enable source separation.[52][53] These ideas can be mirrored at the symbolic level (separate per-instrument bar latents concatenated into joint track-group latents) or used as back-end audio synthesis once a symbolic bar-latent sequence is generated.

Representation and model choices for bar-level latents

Temporal resolution and bar framing

LPD uses 24 time steps per beat, i.e., 96 per 4/4 bar, which covers common 32nd-note and triplet patterns and has been successfully used by multiple generative models.[14][19][13]
SCG's piano-roll encoder uses a 128×128 spatial grid with 10 ms resolution in time, indicating that time resolutions at or above 96 steps per bar are tractable for VAEs and latent diffusion, especially when compressed by a convolutional encoder.[3]

For bar-level latents, prior work suggests several strategies:

- **Fixed-length bar images**: Each bar is rasterized to a fixed H×W×C grid (e.g., 128×96×C or padded/cropped to 128×128×C) so that off-the-shelf image VAEs can be applied.[13][3]
- **Phrase-level windows**: PhraseVAE shows that compressing multiple bars into a single latent yields higher-level phrase semantics but at the cost of per-bar resolution; they also explore per-bar and per-position encodings.[44][7]
- **Bar-wise hierarchical encoders**: MuseMorphose encodes each bar separately with a Transformer encoder, essentially treating each bar as a sequence segment to be mapped into a bar-level latent; such a design is compatible with an image-based front end where each bar-image is first flattened or processed by CNNs.[6]

Channels and semantic axes

Drawing from MIDI-VAE, GrooVAE, and SCG, useful channels for a piano-roll image include:[11][39][20][3]

- Pitch-activity / velocity channel(s): integer or float velocities; or binary with separate learned velocity embedding.
- Onset channel: binary onset-only mask to disentangle note starts from sustained regions (helpful for expressing rhythm and articulations).
- Pedal or sustain channel: for piano and sustaining instruments.[3]
- Track or instrument-type channels: for limited track sets (e.g., {drums, bass, piano, guitar, strings}), use separate channels; or include an instrument-ID mask per pitch-time location if multiple instruments share the same roll.[14][20]

Additional metadata (tempo, key, style, role) can be attached as per-bar/global conditioning vectors (concatenated to the latent or injected via FiLM/attention conditioning) rather than as image pixels.[40][20]

Continuous vs discrete latents

Continuous Gaussian latents are favored in MusicVAE, SCG, PhraseVAE, and Mittal's diffusion-over-latents, providing compatibility with standard VAE and diffusion formulations.[9][26][2][3]
Discrete latents via VQ-VAE, as in PocketVAE and drum-oriented models, are appealing when categorical structure dominates (e.g., drum patterns) and for discrete diffusion methods.[45][25]

A hybrid scheme is also possible: continuous bar-level vectors for pitch/structure and discrete codes for patterns or roles, with both feeding into a conditional diffusion model.

Post-processing and further transformations in latent space

Linear methods: PCA and manifold visualization

Numerous works visualize music VAEs via t-SNE or similar techniques, revealing clustering by style, key, or rhythm, though these are mostly qualitative; PhraseVAE reports exploration of latent smoothness and interpolation behavior as part of its analysis.[26][7]
Linear methods like PCA can be used to identify major directions of variation in learned bar-latent spaces, analogous to the attribute vectors in MusicVAE and tonal-tension VAE.[24][10]

Such analyses can inform:

- Whether the latent space collapses to a small region (posterior collapse).
- Whether certain attributes (density, register, rhythmic complexity) correspond to low-dimensional directions that can be exploited for control.

Metric learning and regularized latent geometries

Deep metric learning vs classification for music representation shows that different training objectives yield different embedding properties: classification-based models often perform better for similarity retrieval and auto-tagging, while metric learning excels at

triplet prediction.[54]
This suggests incorporating contrastive/metric losses alongside reconstruction and KL in VAEs to encourage latent spaces that are useful for downstream similarity and retrieval tasks.

MeasureVAE and GM-VAE-style models incorporate adversarial and mixture priors to enforce disentanglement or to learn clusterable high-level attributes.[47][5]
These approaches can be adapted to bar-level latents to:

- Enforce independence between rhythm, harmony, instrumentation, etc.
- Encourage linear or monotonic relationships between latent dimensions and target attributes.

Evaluation of latent spaces and generative performance

Objective symbolic evaluation metrics

"On the evaluation of generative models in music" proposes a suite of musically informed, reproducible objective metrics for symbolic music, focusing on basic technical properties like pitch-class histograms, tonal tension, rhythmic patterns, note density, and repetition, among others.[33]
The associated **MGEval** toolbox implements these metrics to compare generated music against a reference dataset, providing both absolute and relative measures.[55]

A recent survey on evaluation metrics for music generation reviews both audio and symbolic metrics, distinguishing between reference-based measures (e.g., statistical similarity to training data) and reference-free measures (e.g., internal consistency, structure, diversity), and highlights the relative immaturity of symbolic evaluation compared to audio.[56]

Other work (e.g., MT-GPT-2) proposes domain-specific symbolic metrics (MEM) based on mathematical statistics, music-theory rules, and signal-processing-based features to assess melody quality,

showing that such composite metrics can correlate with expert judgments.[57]

Evaluation of disentanglement and controllability

EC2-VAE, MeasureVAE, and GM-VAE papers design custom metrics for disentanglement and attribute independence, such as measuring how much pitch changes affect rhythm latents and vice versa, or how independent latent dimensions are with respect to labeled attributes.[5][47][27]
Recent audio-focused work evaluates disentangled representations for controllable music generation across axes of informativeness, equivariance, invariance, and disentanglement, using probing models and controlled inputs, and finds that actual embeddings often deviate from intended semantics.[58]

These frameworks can be adapted to symbolic bar-latent spaces by:

- Training linear or shallow probes to predict bar-level attributes (density, tonal center, syncopation) from latents.
- Measuring equivariance/invariance under known transformations (e.g., transposition, time stretching within bar) and checking whether corresponding latent transformations are linear or localized.

Diffusion-specific evaluation

Symbolic diffusion models (Mittal et al., SCG, discrete DDPMs) generally evaluate along:

- **Reconstruction and negative log-likelihood** in latent or token space.[12][9]
- **Objective symbolic metrics** (as above) comparing generated vs training distribution.[9][3]
- **Structure and control metrics**, such as adherence to chord progression or rule-guidance conditions.[59][60][3]
- **Human listening tests** for musicality, coherence, and controllability.[9][3]

These should be included in any evaluation of a bar-latent diffusion pipeline used for symbolic music.

Datasets and tooling

Several datasets and tools are particularly relevant for bar-level latent research:

- **Lakh MIDI Dataset (LMD)** and **Lakh Pianoroll Dataset (LPD)**: large-scale multi-genre symbolic datasets with multi-track piano-roll derivatives; LPD provides efficient storage and 5- or 17-track merged instrument categories suitable for bar-level piano-roll experiments.[61][19][14]
- **Groove MIDI Dataset**: professional drummer performances with precise timing and velocities, used by GrooVAE for expressive drum modeling; ideal for experiments involving groove channels or drumming tracks.[39][11]
- **Pop909 and Pop1k7**: pop-piano datasets used in SCG's high-resolution latent diffusion experiments.[3]
- **Pypianoroll**: core toolkit for converting MIDI to multitrack piano-rolls, trimming bars, merging tracks, and computing tonal distances and other features.[16][62][15]

Additionally, surveys on deep learning for symbolic music generation provide broad overviews of representations (piano-roll, event-based, text-like), architectures (RNNs, Transformers, GANs, VAEs, diffusion), and evaluation methods, which can serve as background reading.[63][64][65]

Suggested pipeline architecture for bar-level latent spaces

Based on the surveyed literature and the desire to reuse pretrained image VAEs, a practical pipeline for the target research could have the following stages.

1. Symbolic preprocessing and piano-roll construction

- Convert MIDI to multitrack piano-rolls using Pypianoroll or a custom pipeline, aligning to a fixed symbolic resolution (e.g., 24 or 32 steps per beat).[15][13]
- Segment into bars, with consistent time signatures (4/4 initially), optionally grouping tracks by instrument family (e.g., drums,

bass, harmony, melody) using LPD's mapping.[66][14]
- For each bar-track or bar-track-group, build an image tensor of shape H×W×C; choose H (e.g., 84 or 128 pitches), W (e.g., 96 or 128 time steps), and channels as:
  - C1: velocity or binary note presence.
  - C2: onset mask.
  - C3: pedal or articulation (if present).
  - Optionally, additional channels for instrument classes or per-track roles, later mapped into RGB or multi-channel images compatible with the chosen VAE.[20][3]

2. Encoding bars with a pretrained image VAE

- Select a pretrained image VAE backend (e.g., SD v1.4 VAE, SDXL VAE, sd-vae-ft-mse, eq-vae-ema, FLUX Tiny AutoEncoder) based on throughput and desired latent resolution.[31][30]
- Define a deterministic mapping from piano-roll channels to the VAE's input channels (typically RGB), and resize/pad bar images to the input resolution required by the VAE (e.g., 128×128, 256×256).
- Run each bar-image through the frozen encoder to obtain a latent tensor (e.g., 16×16×4); optionally apply global average pooling or shallow conv/MLP layers to compress this tensor into a single latent vector per bar per track.
- Optionally, fine-tune the VAE on piano-roll images with mixed image/music losses to better respect musical structure, inspired by Stable Audio's use of domain-appropriate reconstruction losses.[1][29]

If MIDI-specific VAEs are also available, this stage can be run in parallel, providing both a music-specific latent and an image-VAE latent for comparison and possible joint training.

3. Sequencing and hierarchical modeling

- Build sequences of bar-latents across the piece per track and optionally stack them across tracks, akin to MusicVAE's hierarchical conductor and PhraseVAE's latent sequences.[7][26][2]
- For long sequences, consider a hierarchical scheme where phrase-level latents summarize groups of bars, with diffusion or

transformer models operating at the bar level and separate models at the phrase level.

4. Diffusion over latents

- For continuous latents, adopt a DiT/UNet-style diffusion model over sequences of bar-latent vectors, in line with Mittal et al. (diffusion over VAE latents) and SCG's latent diffusion for piano rolls.[9][3]
- Alternatively, train a latent LDM at the phrase level using PhraseVAE-style 64D phrase latents, and use a secondary bar-level VAE (image-based or MIDI-specific) only for reconstruction and editing.[26][7]
- For discrete-coded variants, replace the VAE with a VQ-VAE and use discrete diffusion (e.g., D3PIA, discrete DDPM) over sequences of bar codes.[25][12][45]

Conditioning and control can be integrated via:

- Classifiers or attribute regressors over bar-latents, for classifier-free or SCG-style rule guidance.[23][3]
- Cross-attention to chord graphs, text prompts, or structural templates (as in JASCO, GETMusic, or MuseMorphose).[59][49][6]

5. Multi-modal and feature-augmented extensions

- Attach per-bar side features (instrument IDs, structural role, tonal center, groove features), trained via auxiliary losses to encourage disentangled factors.[47][27][20]
- Map audio and symbolic segments into a joint latent space with a cross-modal encoder (e.g., audio mel-spectrogram encoder and symbolic/image VAE sharing a latent), following audio-to-symbolic arrangement and JASCO.[49][48]
- Use the joint latent as conditioning for both symbolic diffusion and MIDI-to-audio synthesizers like MIAO to end-to-end generate audio from bar-latent sequences.[51][50]

Concluding remarks

Existing literature already provides strong precedents for bar- and phrase-level latent representations of symbolic music, both with sequence VAEs (MusicVAE, EC2-VAE, MuseMorphose, MeasureVAE) and with image-like VAE front ends used for latent diffusion (SCG, PhraseVAE/PhraseLDM).[7][2][5][26][6][3]
By combining piano-roll image encodings with **pretrained image VAEs** (from Stable Diffusion/SDXL-style models), VAE-based bar-latents, hierarchical sequencing, and diffusion over these latents, a flexible and controllable symbolic music generation pipeline can be constructed that aligns well with current state of the art while reusing mature image-latent tooling.[1][30][31][3]
Multi-modal extensions and structured/disentangled latent designs, as demonstrated in MIDI-VAE, GrooVAE, cross-modal arrangement, and recent evaluation work on disentangled music representations, offer clear avenues for incorporating instrument, timbre, audio, and higher-level musical attributes into the same latent framework.[22][58][48][49][11][27][20]

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